ENGINEERING GEOLOGY

UNIT - IV : EARTH HAZARDS & GEOPHYSICS

Ground Water, Earthquakes, Landslides & Geophysical Methods

Syllabus (as given)

- Ground Water: water table, cone of depression, geological controls, exploration techniques. Earthquakes and Landslides: terminology, classification, causes, effects, shield areas and seismic belts, Richter scale, intensity, precautions in seismic areas; classification, causes, effects and prevention of landslides. Geophysics: importance, classification, gravity, magnetic, electrical, seismic, radiometric methods; electrical resistivity, seismic refraction; engineering properties of rocks.

1. Ground Water

Ground water = water present in the **pores, cracks and voids of soils and rocks below the earth's surface**. It is the largest source of fresh water on land (after glaciers) and supplies > 60% of irrigation and drinking water in India.

- **Sources (origin): meteoric** (rain & snow infiltrating), **connate** (trapped in sediments during deposition), **juvenile / magmatic** (from magma), **recycled** (rivers, lakes, canals).
- **Hydrological cycle:** evaporation -> condensation -> precipitation -> infiltration / run-off -> ground water -> springs and rivers -> sea.

1.1 Important Terms

- **Porosity (n)** = volume of voids / total volume × 100 - ability to **store** water. Clay 45-55%, sand 30-40%, sandstone 5-25%, granite < 1%.
- **Permeability (K)** = ability to **transmit** water. Gravel and coarse sand high; clay and shale (high porosity but **very low permeability**).
- **Specific yield** = water that drains by gravity; **Specific retention** = water held by molecular forces. Porosity = Sy + Sr.
- **Aquifer** - rock / soil that stores and **yields water in useful quantity** (sand, gravel, jointed basalt, weathered granite, cavernous limestone).

- **Aquiclude** - stores water but does not transmit (clay); **Aquitard** - poorly permeable (silty clay); **Aquifuge** - neither stores nor transmits (massive granite).
- **Unconfined (water-table) aquifer** - top boundary is the water table, open to the atmosphere. **Confined (artesian) aquifer** - sandwiched between two impermeable layers; water under **pressure**.
- **Perched water table** - local water table above the main one, held on a lens of clay.

1.2 Water Table and Zones

- **Water table** = the **upper surface of the zone of saturation** at which water pressure equals atmospheric pressure. It is **not horizontal**: it **follows the topography** in a subdued form (higher under hills, lower in valleys).
- **Zone of aeration (vadose zone)** - above the water table: soil water belt, intermediate belt, capillary fringe; pores contain **air + water**.
- **Zone of saturation (phreatic)** - below the water table; all pores filled with water.
- **Fluctuation:** rises in monsoon (recharge), falls in summer (discharge, pumping, evaporation). Depth measured in open wells / **piezometers**.
- **Gaining stream** - river fed by ground water (water table slopes to river); **Losing stream** - river recharges the water table.

1.3 Cone of Depression

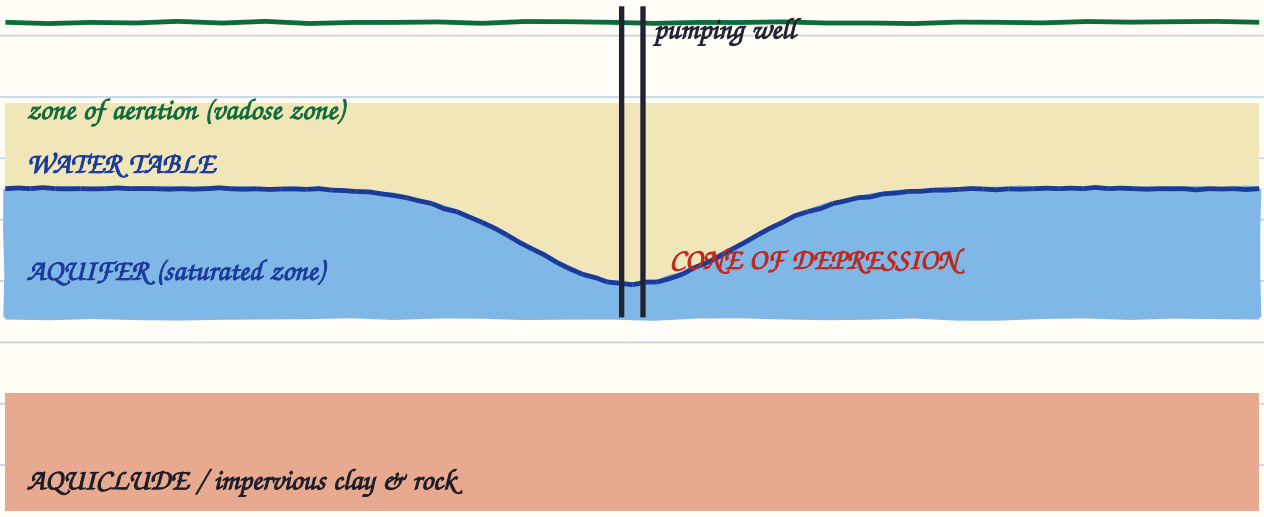


Fig 1: Water table and cone of depression around a pumping well

- When a well is **pumped**, the water level in the well drops and a **cone-shaped depression** forms in the water table around the well - the **cone of depression**.
- **Drawdown (s)** = lowering of water level (static level - pumping level). **Radius of**

influence (R) = distance from the well where drawdown becomes zero.

- Size depends on: **pumping rate**, aquifer **permeability / transmissivity**, thickness, pumping time. Fine-grained aquifer -> **steep, narrow cone**; coarse aquifer -> **gentle, wide cone**.
- **Interference:** cones of nearby wells **overlap**, increasing drawdown and well failure - wells must be **spaced** adequately.
- Over-pumping -> **declining water table, dry wells, land subsidence, sea-water intrusion** in coastal areas (e.g., Krishna / Godavari deltas, A.P. coast).
- Discharge from a well (steady flow): $Q = \pi K (H^2 - h^2) / \ln (R / r)$ for unconfined aquifer (Dupuit's formula).

Remember (diagram!)

- Draw: ground surface, water table, pumping well, drawdown curve, cone of depression, radius of influence, aquifer and aquiclude.

1.4 Geological Controls of Ground Water Movement

Movement and storage depend on the **geological setting**:

- **Lithology (rock type):**
  - **Sedimentary:** sandstone, gravel, alluvium - good aquifers; shale, clay - barriers; **limestone** - karst conduits.
  - **Igneous & metamorphic:** hard crystalline rocks have **negligible primary porosity** - water in **joints, fractures, weathered zone**. **Vesicular basalt, jointed basalt (Deccan)** yield water.
- **Porosity, permeability, grain size & sorting:** well-sorted coarse material = high yield; poorly sorted = low.
- **Structure:**
  - **Folds:** synclines collect water (artesian conditions); anticlines drain.
  - **Faults:** may act as **conduits** (breccia) or **barriers** (gouge); springs occur along faults. **Joints** give secondary porosity.
  - **Dykes** (impermeable) divide aquifers and form **dyke-ponded water**; **unconformities** act as flow paths.
  - **Dip of beds:** water moves along the dip.
- **Topography & drainage:** gentle slopes and valleys favour recharge; steep slopes give

run-off.

- **Weathering:** weathered zone (up to 20-30 m) of granite / gneiss is the **shallow aquifer** in Rayalaseema.
- **Climate & vegetation** control recharge.
- **Springs:** natural outflow where the water table meets surface - **contact springs, fault springs, depression springs, artesian springs.**

1.5 Ground Water Exploration Techniques

1. **Geological methods:** study of **rock types, structures, lineaments, geomorphology, springs, vegetation; geological map** and well inventory. Best sites: **valley fills, alluvium, intersection of fractures (lineaments), contact of dyke and country rock.**
2. **Remote sensing & GIS:** satellite images and aerial photos - **lineaments, drainage pattern, land use**, geomorphic units; used by NRSC for ground-water prospect maps.
3. **Geophysical methods (surface):**
   - **Electrical resistivity (VES)** - most widely used: low resistivity = clay or saturated zone / saline; high = dry rock. **Wenner & Schlumberger** arrays.
   - **Seismic refraction** - depth to bed rock / water table; **Gravity / magnetic** - buried valleys, dykes; **Ground-penetrating radar**, EM methods.
4. **Test drilling and well logging: exploratory bore-holes, electrical logs** (resistivity, self-potential, gamma log), **pumping tests** to find yield and transmissivity.
5. **Hydro-geochemical & isotope methods:** water chemistry (TDS, EC, ions), tracers for age and flow direction.
6. **Water divining / dowsing** - non-scientific.

Practical tip

- In granite-gneiss terrain (Anantapur-Kurnool belt) bore wells succeed along **fractures and lineaments**, not on massive rock; ideal depth 60-100 m (weathered zone + fractured zone).

2. Earthquakes

An **earthquake** is **sudden shaking / vibration of the earth's crust** due to release of strain energy stored in rocks, which travels as **seismic waves**.

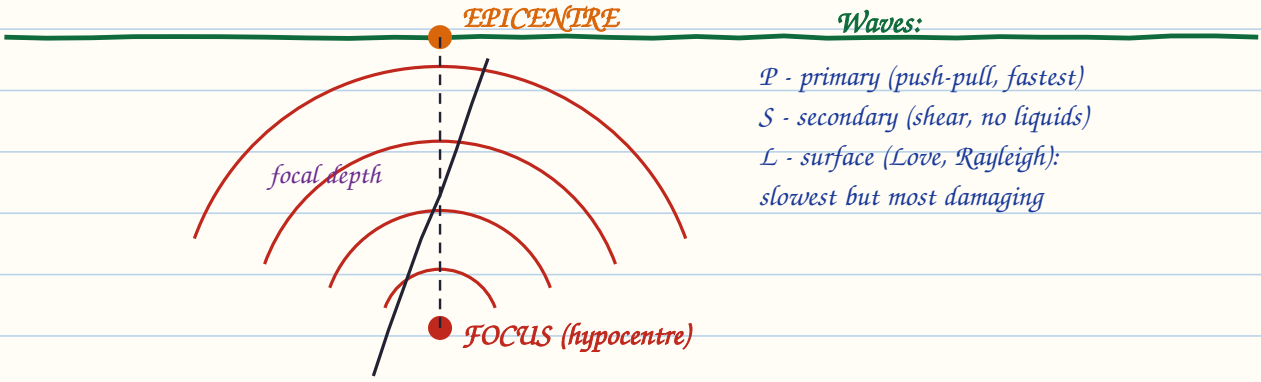


Fig 2: Focus, epicentre and seismic waves

2.1 Terminology

- **Focus (hypocentre)** - point **inside the earth** where rupture starts; **Epicentre** - point on the surface **directly above** the focus.
- **Focal depth:** shallow (< 70 km), intermediate (70-300), deep (300-700 km). Shallow quakes are most destructive.
- **Seismograph** records waves as **seismogram**; **Seismology** = science of earthquakes; **Isoseismal lines** - join places of equal intensity; **Aftershocks / foreshocks.**
- **Body waves: P-waves** (primary, compressional, **fastest**, pass through solids, liquids, gases); **S-waves** (secondary, shear, **only solids**, 1.7 × slower).
- **Surface waves: Love** (horizontal shear) and **Rayleigh** (rolling) - slowest, **cause maximum damage**.
- **S-P time lag** gives distance of epicentre; three stations -> **triangulation** locates epicentre.

2.2 Classification

- **By cause: Tectonic** (fault movement - most common, 90%), **volcanic**, **collapse** (caves, mines), **explosion** (nuclear, blasts), **reservoir-induced** (Koyna 1967, M 6.5).
- **By focal depth:** shallow, intermediate, deep.
- **By magnitude:** micro (< 3), minor (3-3.9), light (4-4.9), moderate (5-5.9), strong (6-6.9), major (7-7.9), **great (≥ 8)**.

2.3 Causes of Earthquakes

- **Plate tectonics** - movement at **convergent, divergent and transform boundaries**; **elastic rebound theory** (H. F. Reid) - rock strains, then snaps along a fault releasing energy.
- Volcanic eruptions, **isostatic adjustment**, landslides, **reservoir impoundment**, mining,

nuclear blasts, **fluid injection**.

2.4 Effects of Earthquakes

- **Primary:** ground shaking, **surface rupture**, **tilting & subsidence**, **liquefaction** (saturated sand behaves as liquid), **landslides**.
- **Secondary:** **collapse of buildings, bridges, dams**, fires, floods, **tsunami** (sea-floor quakes - Indian Ocean 2004), **damage to pipelines**, loss of life, disease.
- Famous Indian quakes: **Latur 1993 (M 6.2)**, **Bhuj 2001 (M 7.7)**, **Uttarkashi 1991**, **Kashmir 2005**, **Sikkim 2011**, Assam 1950.

2.5 Shield Areas and Seismic Belts

- **Shield** = ancient, **stable crust** of Precambrian crystalline rocks; **low seismicity**. **Peninsular India** (Dharwar, Bastar, Singhbhum cratons) is a stable shield - **but not quake-free** (Latur, Koyna, Jabalpur, Bhadrachalam).
- **Seismic belts (zones):** (i) **Circum-Pacific belt** ("Ring of Fire", ~ 80% of energy), (ii) **Alpine-Himalayan (Mediterranean-Trans-Asiatic) belt** ~ 15%, (iii) **Mid-Atlantic ridge / mid-oceanic ridges**.
- **Seismic zones of India (IS 1893:2016): Zone II** (low), **III** (moderate), **IV** (severe), **V** (very severe: Himalaya, NE India, Kutch, Andaman). Zone I does not exist now.
  - **Andhra Pradesh** lies mostly in **Zone II and III** (Kurnool, Anantapur - Zone II/III).

2.6 Magnitude and Intensity

Feature	Richter scale (magnitude)	Mercalli scale (intensity)
Measures	**Energy released** at the focus	**Effects / damage** at a place
Type	Quantitative, logarithmic, instrument-based	Qualitative, I - XII, observation
Variation	One value for a quake	Varies with distance and ground
Meaning	Each **+1 = 10 times amplitude, ≈ 32 times energy**	I = not felt ... **XII = total destruction**

- **Richter (1935):** $M = \log A$ + correction; $M < 2$ not felt; 5 - damage to weak buildings; 7 - serious damage; > 8 - great. Now **moment magnitude (Mw)** is preferred for large quakes.
- **Modified Mercalli (MMI):** I (instrumental) - III (feel indoors) - V (waking) - VII

(damage to poor buildings) - IX (heavy damage) - XII (total).

2.7 Precautions for Building Construction in Seismic Areas

- **Site selection:** avoid **active faults, loose sand (liquefiable), soft clay, steep slopes, filled ground**; prefer **rock or firm soil**.
- **Foundation:** uniform and on **same type of soil**; **raft / strip with tie-beams**; deeper than weathered zone; avoid partial pile / footing mix.
- **Configuration: simple, symmetrical, regular** plan; avoid soft storeys, projections, re-entrant corners, heavy top, large openings; **adequate separation (expansion) joints** between buildings.
- **Materials & detailing:** use **light weight** materials; **ductile detailing (IS 13920)**; strong column - weak beam; **shear walls**, bracing; **bands** (plinth, lintel, roof) and vertical bars in masonry (**IS 4326**).
- **Design:** use **IS 1893** zone factors (Z); **base isolators, dampers.**
- **Special structures:** dams - **flexible earth-fill or gravity dams on rock**, seismic coefficient method; tall chimneys, bridges with **seismic arrestors**.
- Public: **early warning, drills, retrofitting** of old buildings.

3. Landslides

Landslide = the **downslope movement of a mass of rock, debris or earth under the influence of gravity. Mass wasting** is the general term (includes creep, flows, falls).

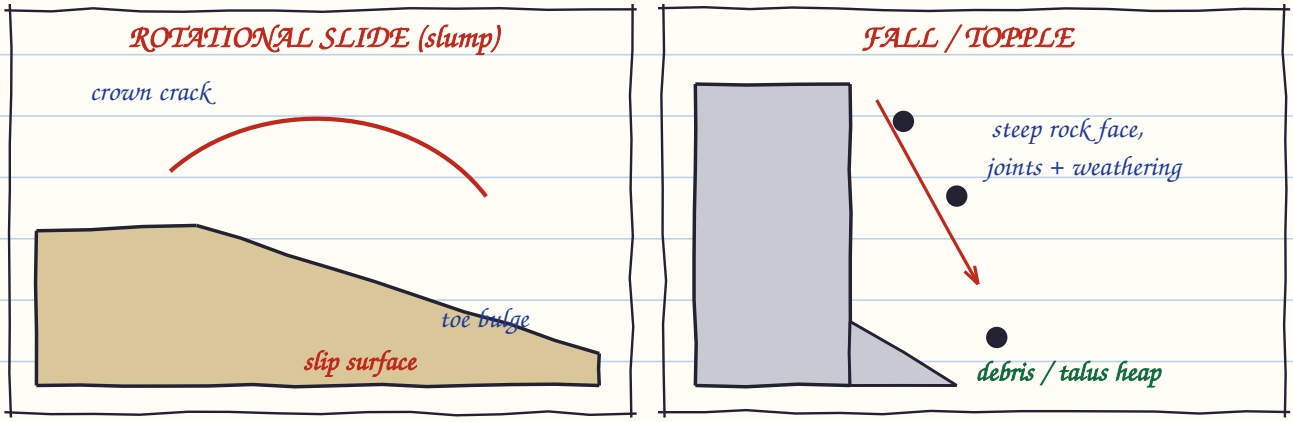


Fig 3: Rotational slide and rock fall

3.1 Classification of Landslides (Varnes / IS 14680)

Type	Description
Falls	Free fall / bouncing of rock and debris from **steep cliffs**; sudden; rock fall, debris fall
Topples	**Forward rotation** of rock columns about a pivot point
Slides	Movement on a **surface of rupture. Rotational (slump)** - curved surface, in homogeneous soil; **translational / planar** - along bedding, joint or fault plane; **wedge** slide - along two intersecting planes
Flows	Material behaves like a fluid - **debris flow, mud flow, earth flow, avalanche**; with heavy rain
Creep	Very **slow** continuous movement of soil, shown by tilted poles, curved tree trunks
Lateral spreading	Extension of soil mass over soft layer - in earthquakes (liquefaction)
Complex	Combination of two or more types

- Also classified by **speed** (very slow < 16 mm/year to extremely rapid > 5 m/s), **material** (rock, debris, earth) and **water content**.

3.2 Causes of Landslides

- **Natural causes:**
  - **Gravity** acting on **steep slopes**; **Heavy rainfall / snow melt** - raises pore pressure, adds weight, reduces shear strength (monsoon slides in Western Ghats, Himalaya).
  - **Geology: weak rocks** (shale, clay, schist, phyllite), **weathered rock, jointed and sheared rock, bedding / joints dipping out of the slope** (daylighting), presence of faults, alternate hard-soft beds.
  - **Earthquakes** and volcanic shocks; **erosion of toe** by river / sea waves; **ground water** rise.
- **Human (anthropogenic) causes:**
  - **Excavation / cutting at the toe** of slopes for roads and buildings (over-steepening); **loading at the top** (buildings, dumps); **deforestation**;
  - **Blasting, mining, quarrying**; **leakage from canals, pipes, reservoir drawdown**; **improper drainage and irrigation.**

3.3 Effects of Landslides

- Loss of life and property; **blocking of roads, railways, rivers** (landslide dams -> floods when breached).
- **Damage to dams, reservoirs, canals, power lines** - sudden waves (Vajont).
- Loss of **agricultural land**, forests, soil fertility; **siltation** of reservoirs; **economic loss**, disruption of transport, tourism.
- Examples in India: **Malpa (1998), Kedarnath (2013), Malin (Pune, 2014), Wayanad (2024), Himalayan highways yearly**.

3.4 Preventive and Control Measures

- **Hazard mapping & zonation** (BIS, GSI) - avoid construction in high-hazard zones; **monitoring** (inclinometers, extensometers, rain gauge, **early warning**).
- **Surface and sub-surface drainage: catch-water drains**, lined side ditches, **horizontal drains (weep holes)**, drainage galleries - the **most effective** measure.
- **Slope modification: reduce slope angle**, **benching / terracing**, remove top load (**unloading**), **fill at the toe (counterweight berm)**.
- **Retaining structures: retaining walls, gabion walls, breast walls, crib walls, soil nailing, rock bolts and anchors, shotcrete, wire-mesh**, piles.
- **Bio-engineering: afforestation**, grass turfing, vetiver, **jute / coir geotextile**.
- **Rock fall control: catch fences, rock sheds, trenches**, scaling of loose rocks.
- **Good practice:** do **not cut slopes at too steep angles**; avoid blasting; **plan roads along stable contours**; **control irrigation and leakage**; proper **spoil disposal**.

Remember (ABC of slope stability)

- A - Angle of slope B - Bedding / joint orientation C - Climate / water -> Reduce driving force, increase resisting force (drain, support, vegetate).

4. Geophysics

Geophysics applies **physics (gravity, magnetism, electricity, elasticity, radioactivity)** to study the **earth's subsurface** from surface (or air / bore-hole) measurements, without direct excavation. **Exploration geophysics** looks for physical-property **contrasts**: density, magnetic susceptibility, resistivity, wave velocity.

4.1 Importance of Geophysical Methods

- **Rapid, non-destructive and cheap** compared with drilling; covers large areas.
- Gives **depth to bed rock**, thickness of overburden, **water table**, fracture / fault zones, **cavities**, buried channels - for **dams, tunnels, foundations, roads**.
- Used in ground water, mineral, oil and gas exploration, archaeology and pollution mapping.
- Limitations: **indirect** (needs interpretation, ambiguity), calibrated with **bore holes**.

4.2 Classification

Method	Property measured	Instrument	Engineering use
Gravity	Variation in **density**, g	Gravimeter	Large structures, caverns, salt domes, ore bodies
Magnetic	Magnetic susceptibility	Magnetometer	Dykes, buried faults, iron ores
Electrical	**Resistivity / conductivity**	Resistivity meter (SP, IP)	Ground water, bed rock depth, cavities
Seismic	**Wave velocity**	Geophones, seismograph	Rock quality, layers, depth to rock
Radiometric	Natural **gamma radiation**	Scintillation counter, Geiger	Uranium, thorium, clay layers, mapping

- **By source: natural source** (gravity, magnetic, SP) and **artificial (active)** (resistivity, seismic). **By location:** ground, airborne, marine, bore-hole.

4.3 Principles of Methods

(a) Gravity Method

- Based on **Newton's law**: $g = G\,M / R^2$. **Denser bodies** (ore, rock) give **positive gravity anomalies**; voids / light rock (salt, caves) give **negative** anomalies.
- Measurement: **gravimeter** at stations; corrections: **latitude, elevation (free-air, Bouguer), terrain, tidal drift**. Units: **mGal**.

(b) Magnetic Method

- Based on **differences in magnetic susceptibility** causing local variations in earth's magnetic field. Magnetite, basic rocks - **high anomalies**; sedimentary - weak.
- **Magnetometer** (proton precession, fluxgate), **unit = nT (nanotesla)**. Corrections:

diurnal variation, IGRF.

(c) Electrical Methods

- **Electrical resistivity method:** current *I* is passed through the ground by two current electrodes, **potential difference *V*** is measured by two potential electrodes. **Apparent resistivity** rho-a = K (V / I); for **Wenner array** rho = 2pi a (V/I).

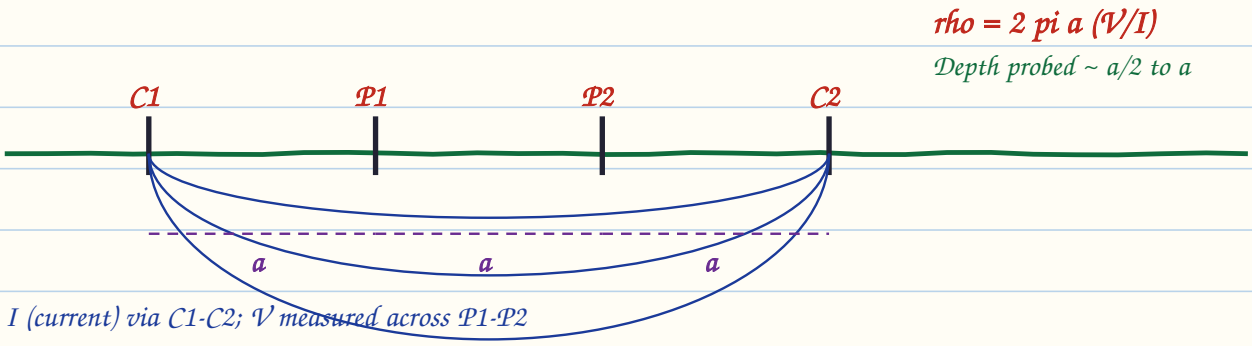


Fig 4: Wenner electrode array

  - **Vertical Electrical Sounding (VES)** - electrodes spacing increased about a centre -> **depth information**; **Profiling** - fixed spacing moved along line -> **lateral changes**.
  - Typical resistivity (Ω·m): **clay 1-20, wet sand 50-200, saline water < 5, weathered rock 20-100, granite 1000+, dry limestone 500-5000.**
- **Self-potential (SP), Induced polarization (IP), EM methods** also used.

(d) Seismic Methods

- Based on **propagation of elastic waves**: artificial source (hammer, explosion), receivers (**geophones**). Velocity increases with **density and rigidity**: soil 200-800 m/s, weathered rock 800-2000, **hard rock 3000-6000 m/s**, water ≈ 1500.
- **Seismic refraction method** - waves refract at interface where **V2 > V1** (critical refraction); gives **depth and velocity** of layers.

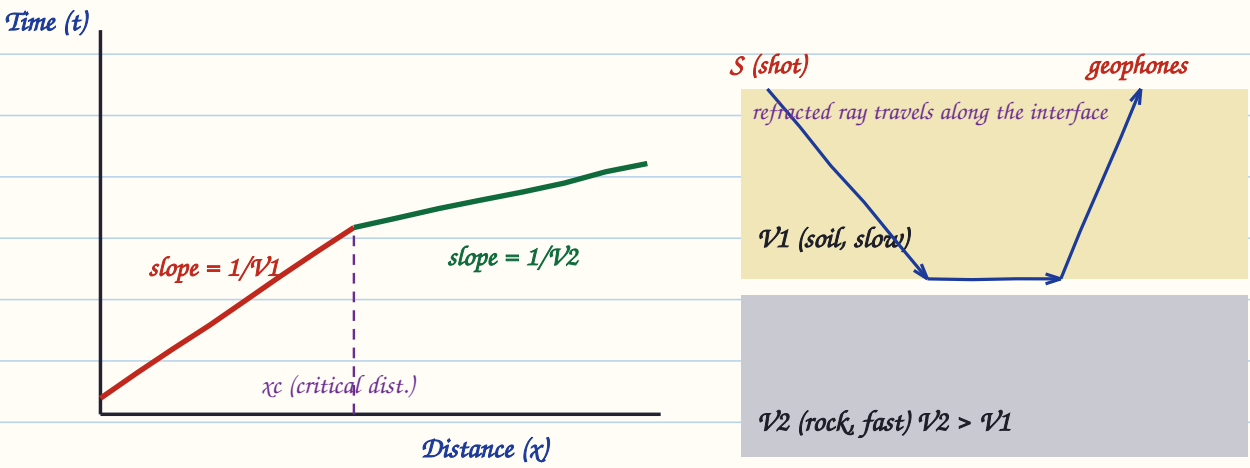


Fig 5: Seismic refraction - time-distance graph

- **Depth of first layer:** $h = (xc / 2) \times \sqrt{[(V2 - V1)/(V2 + V1)]}$ (xc = critical distance) or $h = (ti \times V1\ V2) / (2\sqrt{(V2^2 - V1^2)})$.
- Limitations: **fails if a low-velocity layer lies under a high-velocity layer (blind zone).**

- **Seismic reflection** - echo from boundaries, deep structure (oil exploration).

(e) Radiometric Method

- Measures **natural gamma radiation** from radioactive minerals (**U, Th, K-40**) with **scintillometer / Geiger-Müller counter**; also **gamma-gamma density logging**; used for **uranium exploration** and mapping lithology (e.g., granite, shale).

4.4 Electrical Resistivity vs Seismic Refraction (Quick Comparison)

Aspect	Resistivity	Seismic refraction
Property	Ground resistivity (Ω·m)	Wave velocity (m/s)
Best for	**Ground water, clay layers, saline zones**	**Bed rock depth, rock rippability**
Limitations	Ambiguity from equivalence; metal pipes disturb	Needs velocity increasing with depth; noise
Cost	Low	Moderate

5. Engineering Properties of Rocks

- **Density & specific gravity:** granite 2.6-2.7, basalt 2.8-3.0; **higher density = higher strength**, durability.

- **Porosity & water absorption:** low in igneous rocks (< 1%); high in sandstone and limestone. Good building stone absorption **< 5%** (IS: **< 0.6% for heavy** structures); high absorption -> **frost damage**.
- **Permeability:** primary (pores) and **secondary (joints)** - leakage in reservoirs.
- **Strength: compressive strength** (UCS) is main parameter; tensile strength is only ~ **1/10** of compressive; **shear strength**. **Rock hardness**.
- **Elasticity:** Young's modulus 20-100 GPa for hard rock, Poisson's ratio 0.1-0.3; **elastic, brittle**.
- **Durability / weathering resistance:** resistance to wetting-drying, acids, frost; **slake durability index** - shale low.
- **Abrasion & toughness (Los Angeles abrasion, impact value)** - road aggregate (limit ≈ 30-35%).
- **Anisotropy:** schist, slate, shale behave differently along and across planes.
- **RQD (Rock Quality Designation)** = (sum of core pieces > 10 cm / total core length) × 100.

RQD (%)	Rock quality
< 25	Very poor
25-50	Poor
50-75	Fair
75-90	Good
90-100	Excellent

- **Other rock mass classifications: RMR (Bieniawski), Q-system (Barton)** for tunnels; **Geological Strength Index (GSI)**.

Exam points

- Q: Cone of depression, Richter scale vs Mercalli, landslide control, resistivity method, seismic refraction - each 5-10 marks. Draw sketch + formula + engineering use.

5.1 Short Revision Questions

1. **Define water table.** Upper surface of zone of saturation.
2. **What is an aquifer / aquiclude?** Water-yielding / water-storing but non-transmitting rock.
3. **Define focus and epicentre.** Focus - origin below; epicentre - surface point above it.

4. **Which waves cause maximum damage?** Surface waves (Love, Rayleigh).
5. **Seismic zones of India?** II, III, IV, V.
6. **Name two landslide causes due to man.** Slope cutting, blasting, deforestation, poor drainage.
7. **Principle of resistivity method?** Different rocks have different electrical resistivity; rho = 2pia V/I (Wenner).
8. **Seismic refraction condition?** V2 > V1 (velocity must increase with depth).